

# Yufan Du

---

Email: nbsdyf@stu.pku.edu.cn | GitHub: github.com/Yufan-Du | Website: yufandu.com

## Education

---

**Peking University** **Sep. 2021 - Present**  
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

## Relevant Scores

---

**Overall GPA:** 3.844/4.0 (top 10% of 66 students)  
**TOEFL:** 108(R27 + L30 + S23 + W28)  
**GRE:** 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

## Technical Skills

---

**Knowledgeable with:** C/C++, Python  
**Familiar with:** Python ML libraries (PyTorch, NumPy), Verilog, Embedded systems, FPGA, SQL/Database, JavaScript/HTML, Linux systems, MATLAB, Tcl,  $\text{\LaTeX}$   
**EDA tools:** Familiar with Cadence Innovus, Cadence Virtuoso

## Publications

- 
- **Yufan Du\***, Zizheng Guo\*, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang and Ru Huang (2024). PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning. *2024 Design Automation Conference (DAC24)*.

## Project Experiences

---

**FPGA-Based Snake Game Design Project** **Mar. 2023 - Jun. 2023**  
- Advisor: Professor Xiaohui Duan.  
- Deployed a computer vision algorithm and the classic Snake game on a resource-limited FPGA platform, optimizing resource usage.  
- Developed and implemented a real-time control mechanism for the game using hand movement detection with a camera, enhancing user interaction.  
- Introduced a 3D human-computer interface to improve player engagement and provided valuable insights for IoT edge device development.

**AI-Assisted Schedule Manager** **May. 2023 - July. 2023**  
- Advisor: Professor Wei Guo.  
- Developed a cross-platform scheduling application integrated with AI, applying NLP for user-friendly schedule management with commercial API.  
- Achieved a substantial improvement in user experience by enabling efficient management and reminders through conversational interfaces.

## Research Experiences

---

**Cross-Stage Power Prediction for Integrated Circuits** **Jun. 2023. - Nov. 2023.**  
- Advisor: Professor Yibo Lin.  
- Served as the first author.

- Proposed an innovative VLSI circuit power prediction framework that integrates cross-stage circuit-transformation-aware learning.
- Achieved a significant reduction in error rates and computation time compared to industry-leading commercial tools.
- It thus enabled effective power optimization strategies in early-stage VLSI design.
- *Accepted by DAC 2024.*